

Hierarchical modelling

Bayesian multilevel models

Robbie M. Parks, Elizaveta Semenova

robbie.parks@columbia.edu

Environmental Health Sciences, Columbia University

Aug 14, 2025

Models characterized by several parameters

- In many statistical applications, the model is characterized by several parameters.
- Many real-world problems involve data grouped by source, area, time, etc.
- Non-hierarchical models assume independence between units - often inappropriate.

Outline

- Models characterized by several parameters
- Assumptions of linear models
- What we want beyond linear model
- Hierarchical models
- Levels of hierarchy
- Comparing to frequentist random effects
- Bayesian advantages
- Model selection
- Information criteria

Models characterized by several parameters

- Sample R code:

```
1 data <- read_rds(here("data", "italy", "italy_mortality.rds"))
2 glimpse(data)
3 summary(data)
```

Models characterized by several parameters

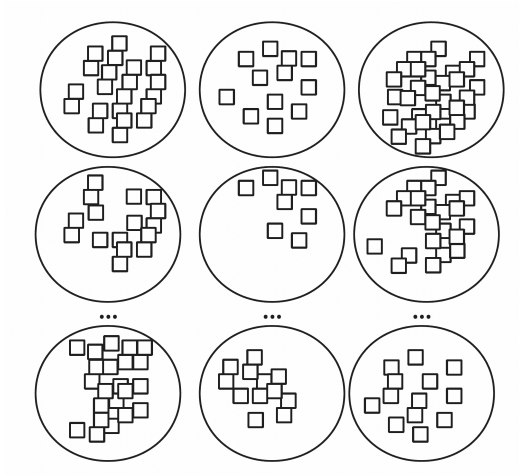
```

Rows: 96,300
Columns: 13
$ NOME_PROVINCIA      <chr> "Agrigento", "Agrigento", "Agrigento",
  "Agrige...
$ SIGLA               <chr> "AG", "AG", "AG", "AG", "AG", "AG", "AG",
  "AG" ...
$ date               <date> 2013-05-04, 2013-05-05, 2013-05-06, 2013-05-
  0...
$ deaths             <dbl> 11, 11, 12, 8, 10, 9, 14, 14, 5, 12, 10, 11,
  9...
$ year               <dbl> 2013, 2013, 2013, 2013, 2013, 2013, 2013,
  2013...
$ expected            <dbl> 11.75367, 11.75367, 11.75367, 11.75367,
  11.753...
$ hol                 <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
  0...

```



Models characterized by several parameters



Models characterized by several parameters

NOME_PROVINCIA	SIGLA	date	deaths
Length:96300	Length:96300	Min. :2013-05-04	Min. : 0.00
Class :character	Class :character	1st Qu.:2014-07-17	1st Qu.: 7.00
Mode :character	Mode :character	Median :2016-01-16	Median : 11.00
		Mean :2016-01-16	Mean : 14.93
		3rd Qu.:2017-07-17	3rd Qu.: 17.00
		Max. :2018-09-30	Max. :272.00

year	expected	hol	month
Min. :2013	Min. : 2.909	Min. :0.00000	Min. :5.000
1st Qu.:2014	1st Qu.: 8.231	1st Qu.:0.00000	1st Qu.:6.000
Median :2016	Median : 11.674	Median :0.00000	Median :7.000
Mean :2016	Mean : 16.223	Mean :0.01333	Mean :7.033
3rd Qu.:2017	3rd Qu.: 18.153	3rd Qu.:0.00000	3rd Qu.:8.000
Max. :2018	Max. :118.261	Max. :1.00000	Max. :9.000

temperature_lag0_3	relativehumidity_lag0_3	o3_lag0_3	pm10_lag0_3
Min. :2.500	Min. :24.00	Min. :1.00	Min. :0.000



Models characterized by several parameters





Assumptions of linear models

- Independence of observations.
- Constant error variance (homoskedasticity).
- Single set of parameters apply to all observations.
- No measurement error in predictors.
- Complete data.



Assumptions of linear models

- Independence of observations. **Data often clustered.**
- Constant error variance (homoskedasticity). **Variability often differs across groups.**
- Single set of parameters apply to all observations. **Subgroups may have different parameters.**
- No measurement error in predictors. **Predictors are often noisy.**
- Complete data. **Missingness is common.**

10

What we want beyond linear model

- To appropriately model grouped or nested data (e.g., individuals within regions, time within subjects).
- To allow for group-specific effects (e.g., varying intercepts or slopes across groups).
- To borrow strength across groups, improving estimates for small or noisy groups.
- To relax the assumption of independence by accounting for intra-group correlation.
- To model heterogeneity in variances across groups or levels.
- To introduce structure in parameters, linking them through shared hyperparameters.

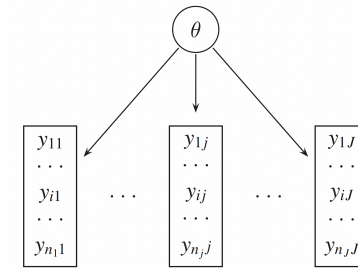
11



What we want beyond linear model

- To propagate uncertainty across levels of the model (data, parameters, hyperparameters).
- To handle missing data naturally through the model hierarchy.
- To incorporate prior knowledge at multiple levels of the model.
- To support flexible generalizations (e.g., nonlinear relationships, spatiotemporal structure, GLMs).
- To incorporate prior information and quantify uncertainty at multiple levels.
- To represent spatial, temporal, or hierarchical dependence in data.

Complete pooling



Complete pooling (from lab)

Priors:

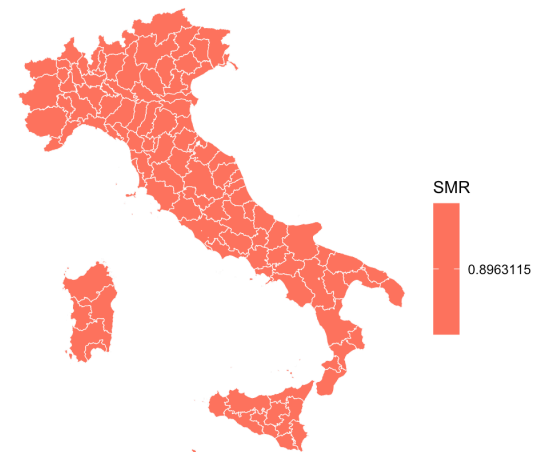
$$\alpha \sim N(0, 5)$$

Likelihood:

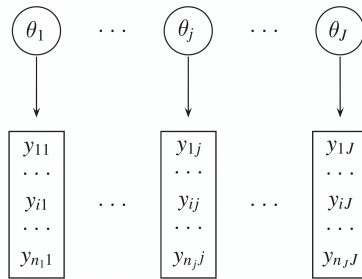
$$y_i \sim \text{Pois}(\mu_i) \quad i = 1, \dots, N$$

$$\log(\mu_i) = \log(E_i) + \alpha$$

Complete pooling (from lab)



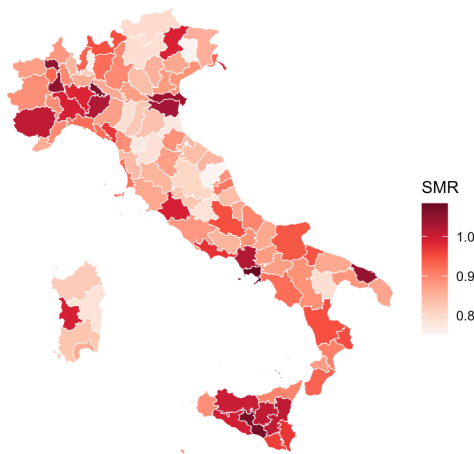
No pooling



- An *unpooled* model implies that the data are sampled from independent parameters.
- $(\theta_1, \theta_2 \text{ etc.})$ drawn independently from e.g., $\theta_j \sim N(0, 1000)$.
- Parameter θ_j is different for each set of observations.
- No exchange of information between θ_j 's.



No pooling (from lab)



No pooling (from lab)

Priors:

$$\alpha_j \sim N(0, 1) \quad j = 1, \dots, N_p$$

Likelihood:

$$y_i \sim \text{Pois}(\mu_i) \quad i = 1, \dots, N$$

$$\log(\mu_i) = \log(E_i) + \alpha_{j[i]}$$



Hierarchical models

- Many real-world problems involve data grouped by source, geography, time, etc.
- Non-hierarchical models assume independence—often inappropriate.
- Hierarchical models:
 - Allow varying group-specific effects.
 - Improve estimates through sharing information (shrinkage).

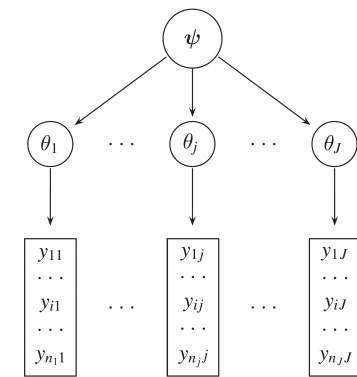


Hierarchical models

Exist on the continuum between two extreme cases:

- Complete pooling (one common parameter).
- **Partial pooling!**
- No pooling (independent parameters).

Partial pooling



- In a *partially pooled* or *multilevel* or *hierarchical* model, parameters are viewed as a sample from a distribution of parameters.



Partial pooling (from lab)

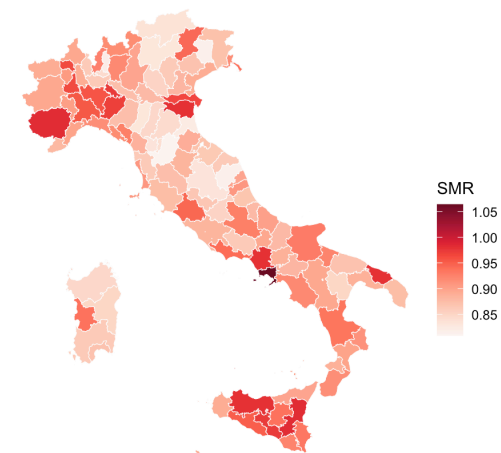
Priors:

$$\begin{aligned}\alpha &\sim N(0, 5), \\ \sigma_p &\sim N^+(1) \\ \theta_j &\sim N(0, \sigma_p^2) \quad j = 1, \dots, N_p\end{aligned}$$

Likelihood:

$$\begin{aligned}y_i &\sim \text{Pois}(\mu_i) \quad i = 1, \dots, N \\ \log(\mu_i) &= \log(E_i) + \alpha + \theta_{j[i]}\end{aligned}$$

Partial pooling (from lab)



Hierarchical (partial pooling) models

- Hierarchical models (partial pooling) fill in the continuum between the two extremes.
- They allow us to estimate models for each measurement block where each dataset is being fit to its own model.
- There is a higher level model, a *hierarchical model*, describing variability in parameters of those sub-models.



Levels of hierarchy

- Data
- Process model (likelihood)
- Parameter model (priors)
- Hyperparameters (hyperpriors)



Comparison of pooling strategies

Pooling Type	Group estimates independent?	Shrinkage	Frequentist equivalent	Bayesian benefit
No Pooling	Yes	None	Fixed effects	None
Full Pooling	No (same for all)	Complete	No group effects	Simple prior
Partial Pooling	No (partially informed)	Adaptive	Random effects	Full posterior with hierarchy



Partial pooling again

Priors:

$$\begin{aligned}\alpha &\sim N(0, 5), \\ \sigma_p &\sim N^+(1) \\ \theta_j &\sim N(0, \sigma_p^2) \quad j = 1, \dots, N_p\end{aligned}$$

Likelihood:

$$\begin{aligned}y_i &\sim \text{Pois}(\mu_i) \quad i = 1, \dots, N \\ \log(\mu_i) &= \log(E_i) + \alpha + \theta_{j[i]}\end{aligned}$$



Comparing to frequentist random effects

- Frequentist random effects models are analogous in some ways to Bayesian hierarchical models, but with important differences.
- They allow for group-specific effects and account for intra-group correlation.
- However, they do not:
 - Provide a full probabilistic framework.
 - Incorporate prior information.



Bayesian advantages

- Handles complex multilevel and spatial hierarchies.
- Uses structured priors (CAR, GMRF) for spatial dependence.
- Supports flexible, non-Gaussian priors.
- Provides full uncertainty via posterior distributions.
- Easily incorporates expert knowledge through priors.
- Enables spatiotemporal and nonstationary extensions.
- Frequentist methods often limited or approximate.



Comparing to frequentist random effects

- Bayesian hierarchical models:
 - Allow for uncertainty quantification at multiple levels.
 - Provide a coherent framework for model comparison and selection.
 - Enable the incorporation of prior knowledge and beliefs.
 - Allow for extensions in complexity on multiple levels (e.g., spatiotemporal models).



Extending hierarchy

- Suppose there's another level of hierarchy, e.g.,
 - another level of geographic region below the regional level (e.g., city),
 - or individuals within regions.
 - We can extend the hierarchy by adding another level of parameters.
- For example, we can add a level for individuals within regions.
- This allows us to model individual-level variability while still accounting for regional effects.



Extending hierarchy

- The model can be expressed as:

$$\begin{aligned} y_i &\sim \text{Pois}(\mu_i) \quad i = 1, \dots, N \\ \log(\mu_i) &= \log(E_i) + \alpha + \theta_{j[i]} + \phi_{k[i]} \\ \phi_k &\sim N(0, \sigma_\phi^2) \quad k = 1, \dots, N_r \end{aligned}$$

- Here, $\phi_{k[i]}$ represents the individual-level effects, and σ_ϕ^2 is the variance of these effects.
- This allows us to capture individual-level variability while still accounting for regional effects.



Information criteria

Information criteria are popular tools for model selection that provide a quantitative measure of the trade-off between model fit and complexity. The *lower* the value of IC, the better.



Model selection

Common information criteria

- AIC (Akaike Information Criterion):

$$\begin{aligned} \text{AIC} &= -2 * \ln(L) + 2k, \\ L &\text{ -- maximum value of likelihood,} \\ k &\text{ -- number of parameters} \end{aligned}$$

- BIC (Bayesian Information Criterion):

$$\text{BIC} = -2 \ln(L) + k \ln(n)$$

- Widely Applicable Information Criterion (WAIC):

$$\text{WAIC} = -2 \sum_{i=1}^n (\ln(\hat{p}(y_i)) - \text{Var}(\ln(\hat{p}(y_i))))$$



Getting ready for the lab

The lab for this session

- This lab will involve introducing hierarchical modelling using the [NIMBLE](#) modelling framework.
- During this lab session, we will:
 1. Write a hierarchical model in [NIMBLE](#);
 2. Compare the model to more basic non-hierarchical models; and
 3. Discuss the advantages of hierarchical models.



Questions?

